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RESEARCH ARTICLE

Dual Gradient Based Snow Attentive Desnowing

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ABSTRACT In this paper, we propose a novel dual gradient-based desnowing algorithm that can accurately remove snow from a scene by characterizing snow particles. To localize snow in an image, we present a gradient-based snow activation map that can be estimated using snow classification. To recognize various patterns in the shapes and trajectories of snow particles, we introduce a gradient-based snow edge map. Using these two gradients, we estimate an accurate snow attention mask that is subsequently used for snow removal. In addition, we propose a translucency-aware context restoration network to handle various degrees of snow transparency and thus, prevent our method from losing the image context information during desnowing. Experimental results demonstrate that the proposed method considerably outperforms other state-of-the-art desnowing algorithms quantitatively and qualitatively.

INDEX TERMS Computer vision, deep learning, image analysis.

I. INTRODUCTION

Severe weather condition changes (*e.g.*, haze, rain, and snowfall) in an outdoor scene typically degrade the quality of images, which significantly worsens the performance of conventional computer vision algorithms, including object detection, segmentation, and visual tracking. To solve this problem, there have been several attempts to deal with various haze and rain conditions [1], [2]. For example, deraining aims to estimate rain streak using deep neural networks, whereas dehazing aims to restore information directly behind the haze. Each task is important to improve visibility of images, thus improve the performance of computer vision applications. While dehazing, deraining, and desnowing have been considered with different characteristics of weather condition, they have the same purpose in terms of restoring the image. However, there are relatively few works on desnowing because it is nontrivial to capture the dynamics of snowing, whereas haze is an atmospheric phenomenon

with only static properties. In addition, estimating the snow particles is more challenging than rain streaks, because snow particles have more diverse shapes and sizes of patterns. Moreover, the snow particles are distributed non-uniformly in an image, whereas rain streaks typically appear more uniform than snow particles. Thus, it is difficult to localize snow. Various degrees of snow transparency also make desnowing significantly challenging. There are difficulties due to the loss of information in the image due to various transparency. Recently, deep learning-based desnowing methods have been proposed [3], [4] and have shown promising performance. Liu et al. [4] presented a snow model using an aberration map. Chen et al. [3] proposed a snow removal method that considers the veiling effect, which occurs if small snow particles are accumulated.

In this paper, in contrast to conventional methods, we represent snow images as a combination of clean and snow mask layers using conventional models of image degradation [5].

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$$S(x) = C(x) + M(x), \quad (1)$$



FIGURE 1. Real-world snow removal results of our proposed method, which can accurately remove snow and make clear images. The first and second rows show input snow images and our desnowing results, respectively.

where $S(x)$ and $C(x)$ denote snow and clean images, respectively, and $M(x)$ denotes a snow mask. Unlike other low-level vision tasks, $M(x)$ is modeled mathematically by reflecting the fact that pattern shape and size of snow particles vary significantly and depend on the shutter speed of the camera. Our method deals with irregular sizes, shapes, and transparency of snow particles using a novel dual gradient-based method, *i.e.*, gradient-based snow activation map (GSAM) and gradient-based snow edge map (GSEM). To estimate the GSAM, we extract the activated gradient map on the snow labels from the target network layers using the gradient-class activation mapping (gradient CAM) [6], for which the network is trained as a binary classification on snow and clean images. This map approximates the snow positions with the corresponding confidence values. To obtain the GSEM, we exploit the edge information using the gradients at the image level. In particular, we obtain this map using hand-crafted convolution filters and a few layers. This map can recognize various shapes and sizes of snow particles. These two gradient-based maps are fed into a mask estimation module to produce the snow attention mask (SAM), $M(x)$. Subsequently, a crude desnowing image can be obtained by just calculating the differences between the observed snow image, $S(x)$, and the snow mask image, $M(x)$, in (1). However, due to various degrees of snow transparency, it is difficult to recover the image context after desnowing. To solve this problem, our method enhances the crude desnowing results using the proposed translucency-aware context restoration network.

The main contributions of this paper are as follows:

- We present a dual gradient-based desnowing method, in which the GSAM localizes snow particles, and the GSEM determines their shapes and sizes.

- We propose a novel mask estimation network that produces the SAM based on two gradient-based maps. In addition, we introduce a translucency-aware context restoration network to recover the information lost owing to opacity and to handle various snow transparency.
- The proposed method typically outperforms other state-of-the-art desnowing methods.

Figure 1 shows snow removal images of our method.

II. RELATED WORKS

A. SINGLE IMAGE DEHAZING AND DERAISING

Dehazing and deraining have been actively studied in the context of image restoration [5], [7]. Yang et al. [5] presented a contextualized dilated network for deraining, which considered the diverse appearances of rain streaks. Chen et al. [7] introduced a context aggregation network to enhance both dehazing and deraining accuracy. In contrast to dehazing, desnowing problems are more challenging because we have to capture various dynamics of snowing. Unlike deraining, desnowing problems are more difficult because snow particles have diverse shapes and sizes; they are distributed non-uniformly in an image, and they have various degrees of transparency.

B. SINGLE IMAGE DESNOWING

There are relatively few works on image desnowing compared to dehazing and deraining. Because snow is a dynamic weather condition, desnowing is typically considered in video processing methods. For example, Nayar and Narasimhan [8] researched the visual manifestations under different weather conditions such as rain and snow. Barnum et al. [9], [10] detected and removed rain and snow particles in the frequency domain of videos using a global effect model. Bossu et al.

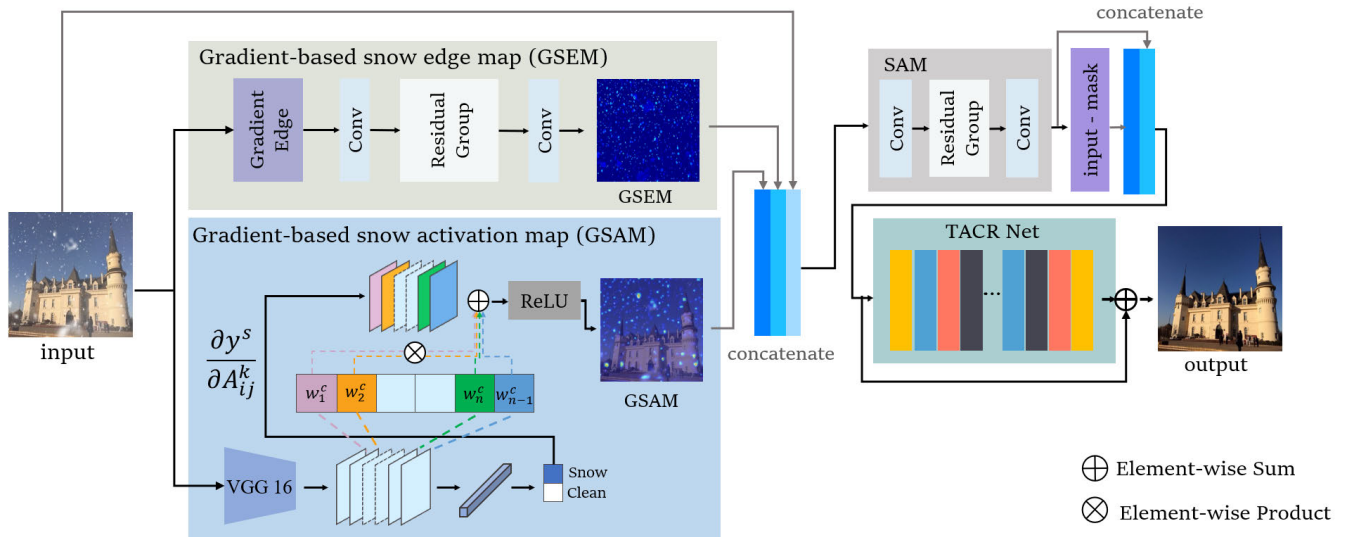


FIGURE 2. Overall framework of our method comprises two gradient-based modules for the snow attention mask (SAM) and translucency-aware context restoration network. Using our dual gradient modules, we estimate an accurate SAM. Context loss caused by snow removal can be restored using translucency-aware context restoration networks.

[11] proposed a rain and snow detection method for videos using a histogram of the orientation of streaks. Unlike the aforementioned video-based desnowing methods that can use snow motion information, single image-based desnowing cannot use this information to remove the snow, which makes a desnowing task challenging. Xu et al. [12] removed rain and snow from images using a single guidance image. Yu et al. [13] proposed a content-adaptive method for rain and snow removal from images based on single-image filtering. Pei et al. [14] proposed a rain and snow removal method for images using saturation and visibility features. Ding et al. [15] removed rain and snow from images using L_0 -smoothing filters. Wang et al. [16] presented a snow removal method for images using image decomposition and dictionary learning. Liu et al. [4] developed a multi-stage network that considered the diversity of translucent or opaque snow particles and restored detailed image information lost by snow particles. Chen et al. [3] proposed a new snow model with the veiling effect and presented a joint size and transparency-aware snow removal algorithm. In contrast, our method introduced a gradient-based desnowing method, which enables the handling of various shapes and trajectories of snow particles based on dual gradients in a single image. Cheng et al. [17] accurately removed snow areas from images using on advanced snow masks. Ye et al. [18] considered the similarity between dehazing and desnowing problems to solve both problems. MSP-Former [19], LMQFormer [20], and SnowFormer [21] employed transformer-based methods, which enabled to produce accurate desnowing results.

C. WEAKLY SUPERVISED OBJECT LOCALIZATION

Our method can accurately determine the snow locations using weakly supervised object localization (WSOL)

algorithms. The WSOL method aims to determine the object locations using only lass labels, without any expensive bounding box or segmentation mask information. For this purpose, Zhou et al. [22] estimated class activation maps using global average pooling. Selvaraju et al. [6] proposed gradient-weighted class activation maps (Grad-CAM) by highlighting the important regions in the image using the gradient of target labels with respect to the target layers in a classification network. Singh and Lee [23] intentionally occluded certain parts of image patches to enhance the classification performance of target class regions. Zhang et al. [24] proposed an adversary classifier that removed activated regions with high scores and retrained the network. In contrast, our method estimated the exact positions of snow particles using the proposed gradient-based WSOL algorithms.

III. PROPOSED METHOD

To handle complex snow shapes and various sizes, we introduce a snow attention mask using gradient-based snow activation map and gradient-based snow edge map in Section III-A. In addition, to recover the information missing due to the translucency or opacity of snow particles, we present a translucency-aware context restoration network in Section III-B. We describe the detailed loss functions and network architectures in Section III-C. Figure 2 shows the overall pipeline of the proposed method.

A. SNOW ATTENTION MASK ESTIMATION

1) GRADIENT-BASED SNOW ACTIVATION MAP (GSAM)

We accurately determine snow positions using the proposed GSAM, which is estimated based on WSOL algorithms (e.g., gradient CAM [6]). In particular, we train a binary classification network to classify the snow using the training

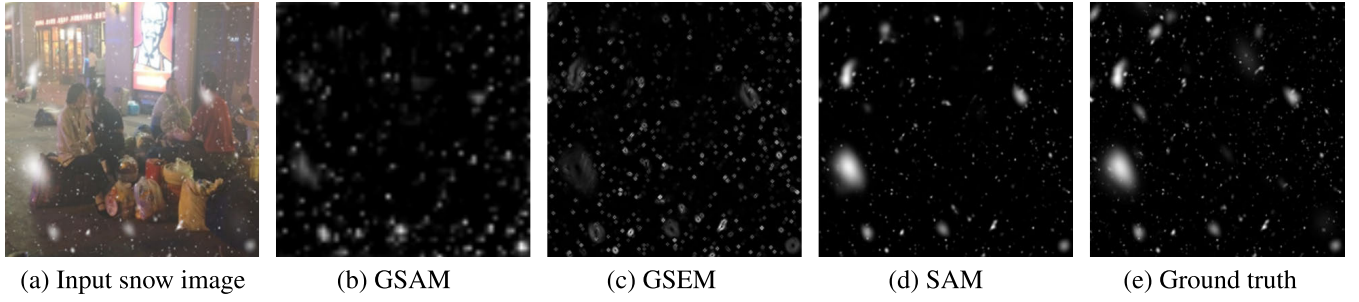


FIGURE 3. Visualization of each component of the proposed method From left to right: input snow image, gradient-based snow activation map (GSAM), gradient-based snow edge map (GSEM), snow attention map (SAM), and ground truth. Our estimated SAM image is extremely similar to the ground truth.

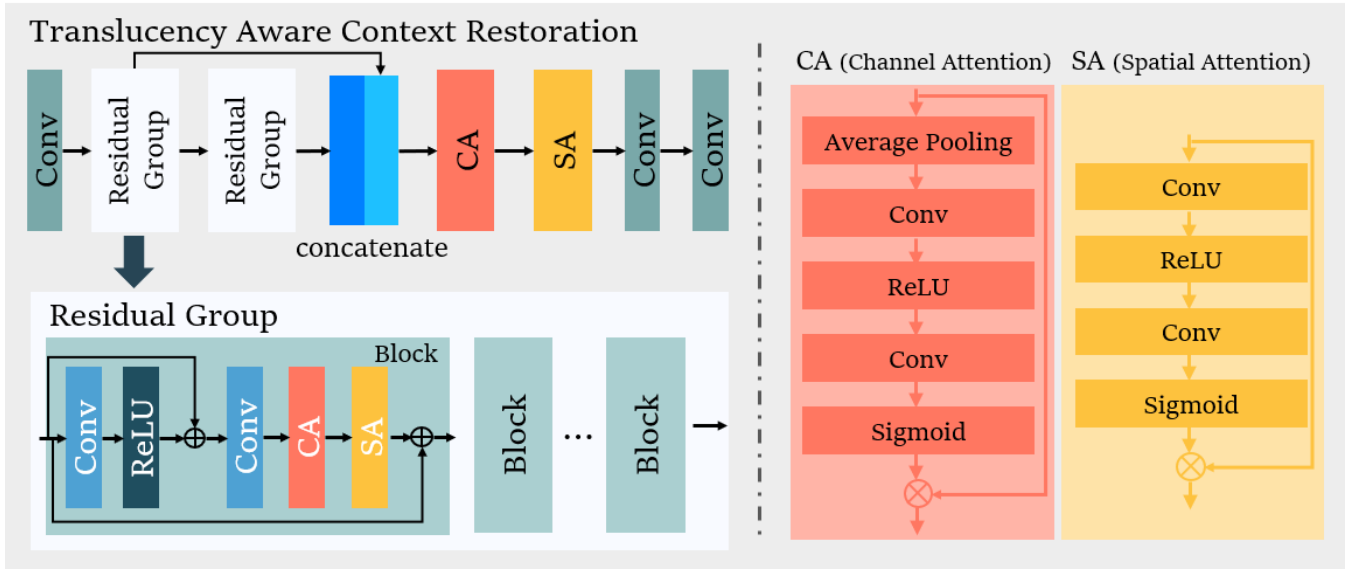


FIGURE 4. Detailed illustration of the proposed translucency-aware context restoration (TACR) network that restores image context information after attaining SAM.

data, $x \in \mathbb{R}^{240 \times 240 \times 3}$, and corresponding labels, $y \in \{0, 1\}$. Subsequently, we calculate the gradient of labels with respect to the target convolutional layers.

$$\alpha_k^s = \frac{1}{W \times H} \sum_i \sum_j \frac{\partial y^s}{\partial A_{ij}^k}, \quad (2)$$

where y^s denotes the output of the classification network before using a softmax function, corresponding to features of snow labels, and A_{ij}^k denotes a target convolutional layer. We use a conv3-3 ReLU layer as the target layer because it includes the semantic information on snow with a sufficiently large feature map. In (2), α_k^s is obtained using the global average pooling of the gradients. Subsequently, the proposed GSAM is estimated as follows.

$$L_{GSAM}^s = \text{ReLU} \left(\sum_k \alpha_k^s A^k \right), \quad (3)$$

where L_{GSAM}^s is normalized and upsampled to obtain a snow mask with the original image size.

2) GRADIENT BASED SNOW EDGE MAP (GSEM)

We consider the complex shapes and various sizes of snow particles using the proposed GSEM, which is estimated by Sobel filters [25]:

$$\nabla = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right], \quad (4)$$

and

$$\frac{\partial f}{\partial x} = S_x \otimes f, \quad \frac{\partial f}{\partial y} = S_y \otimes f, \quad (5)$$

where S_x and S_y denote Sobel kernels for x and y coordinates, respectively, and f denotes an input image. Like [26] that uses a fixed kernel to generate details of high-resolution, this fixed kernel can easily extract high-frequency information of snow particles. To suppress the edges of other objects and amplify the edges of snow particles, we use residual groups with two convolution layers and three residual blocks.

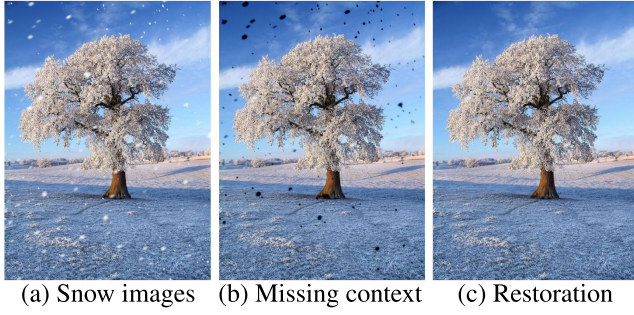


FIGURE 5. Missing context and restored images after removing the snow mask. Missing context images vary according to different degrees of snow translucency. Nevertheless, our method accurately restores the images.

3) SNOW ATTENTION MASK (SAM)

To estimate the exact SAM, we use dual gradient-based methods. These are, as shown in Figure 3(b), a GSAM indicating the approximate position and size of the snow activation, and a GSEM represented as depicted in Figure 3(c), which provides the outline, various sizes, and patterns of snow. The final snow activation map is estimated using the mask estimation module using these dual gradient-based maps and snow images, as shown in Figure 3(d). The estimated snow attention map is similar to the ground-truth snow mask, as depicted in Figure 3(e). The mask estimation module includes two convolution layers and one residual group with three residual blocks.

B. TRANSLUCENCY-AWARE CONTEXT RESTORATION

In (1), we represent the snow model as

$$C(x) = S(x) - M(x), \quad (6)$$

where a clean image, $C(x)$, can be obtained by determining the difference between a snow image, $S(x)$, and a snow mask, $M(x)$, at the x -pixel coordinate. However, in real-world settings, $M(x)$, as estimated in Section III-A, can be easily contaminated with other atmospheric particles [3]. While this problem can be alleviated [3], [4] using inpainting algorithms [27], snow masks typically have non-hole shapes that degrades the inpainting performance. Thus, to solve this problem more explicitly, we redesign $M(x)$ and infer the advanced snow mask, M_{adv} , using our translucency-aware context restoration algorithm, which enables to more accurately restore the image context, $C(x)$.

$$M(x) = M_{adv}(x)(1 - T(x)) + (C(x) + \alpha(x))T(x), \quad (7)$$

where $T(x)$ denotes the translucency function, and $\alpha(x)$ denotes contaminating atmospheric particles. If $T(x) = 0$, $M(x)$ denotes pure snow particles, $M_{adv}(x)$. If $T \neq 0$, $M(x)$ denotes contaminated snow particles combined with the image context and other atmospheric particles. Figure 5(b) shows the image context information lost induced by snow translucency. Our method recovers this missing context using our translucency-aware context restoration network, as shown in Figure 5(c).

To estimate M_{adv} and restore $C(x)$ in (7), we propose a translucency-aware context restoration network based on the networks for super resolution [28] and dehazing [29]. Figure 4 presents our translucency-aware context restoration network comprising two residual groups with 20 residual blocks. The convolution filters used in the residual groups have 48 channels, kernel sizes of 3, and dilation factors of 2. Each residual group has short and long connections for local and global residual learning, respectively, which helps the network to use low-frequency information and focus on the attentive regions. In addition, our restoration network adopts channel and spatial attention. The channel attention (CA) and spatial attention (PA) in right side of Figure 4 perform channel and spatial-wise re-weighted about feature maps. The re-calibrated features using these attention modules can restore $C(x)$ and thus estimate M_{adv} in (7).

Please note that we estimate the snow mask $M(x)$ using (7). For this, we use the ground-truth snow masks during the training. However, because we carefully model $M(x)$ to resemble realistic snow appearances and conditions by considering the snow transparency $T(x)$ and contaminating atmospheric particles $\alpha(x)$, our method can accurately estimate $M(x)$ even in real-world scenarios as well as synthesized scenarios. Figures 1 and 10 demonstrate that our desnowing results are empirically better than those of other methods, because the snow mask $M(x)$ is accurately estimated in real-world scenarios.

C. LOSS FUNCTION

The loss function of our method is defined as follows:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_r + \lambda_2 \mathcal{L}_p + \lambda_3 \mathcal{L}_e + \lambda_4 \mathcal{L}_m + \lambda_5 \mathcal{L}_{refine}, \quad (8)$$

where $\mathcal{L}_r$, $\mathcal{L}_p$, $\mathcal{L}_e$, $\mathcal{L}_m$, and $\mathcal{L}_{refine}$ denote the reconstruction, perceptual, edge, snow mask, and mask edge refine losses, respectively. In (8), hyperparameters $\{\lambda_i\}_{i=1}^5$ are empirically determined.

1) CLASSIFICATION LOSS

To estimate the proposed GSAM, we design the classification loss $\mathcal{L}_b$.

$$\mathcal{L}_b = - \sum_{i=1}^n y^i \log(x), \quad (9)$$

where x denotes the network output after the softmax function and $y = \{0, 1\}$ denotes binary labels. This loss is not included in (8), but used to estimate the GSAM.

2) RECONSTRUCTION LOSS

We make the desnowed images, $G(I)$, close to clean images, C , using $\mathcal{L}_r$.

$$\mathcal{L}_r = \|G(S) - C\|_1, \quad (10)$$

where $G(\cdot)$ denotes the proposed network.

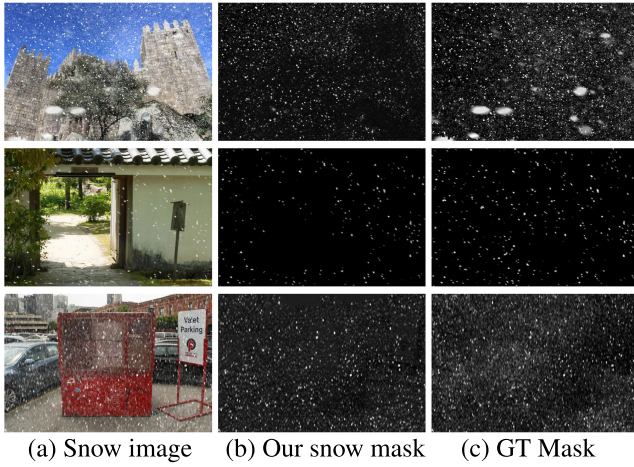


FIGURE 6. Ablation study on qualitative comparison of estimated snow attention masks and GT masks.

3) PERCEPTUAL LOSS

We measure the contextual similarity between $G(S)$ and C at the feature level using L_p .

$$\mathcal{L}_p = \|\phi(G(S)) - \phi(C)\|_1, \quad (11)$$

where $\phi(\cdot)$ estimates the 22-th feature map obtained using the VGG-19 network [30] pretrained using the ImageNet dataset [31].

4) EDGE LOSS

We calculate the shape similarity between objects in $G(S)$ and C using L_e , which enables the removal of snow particles more accurately.

$$\mathcal{L}_e = \|E(G(S)) - E(C)\|_1, \quad (12)$$

where $E(\cdot)$ extracts Sobel edges for three color channels.

5) SNOW MASK LOSS

The SAM is estimated using L_m . To handle various shapes, sizes, and transparencies of snow particles, we use dual gradients.

$$\mathcal{L}_m = \|DG(S) - M_{gt}\|_1, \quad (13)$$

where $DG(\cdot)$ denotes the proposed dual-gradient module and M_{gt} denotes the ground-truth for the snow mask.

6) MASK EDGE REFINE LOSS

Because the Sobel mask is a non-learnable filter, it typically extracts edges with fixed shapes, which can induce several unnecessary edges (*e.g.*, objects). To solve this problem, we compute the edge refine loss.

$$\mathcal{L}_{refine} = \|GE(S) - E(M_{gt})\|_1, \quad (14)$$

where $GE(\cdot)$ estimates the GSEM and $E(M_{gt})$ produces an edge of the ground-truth for the snow mask.

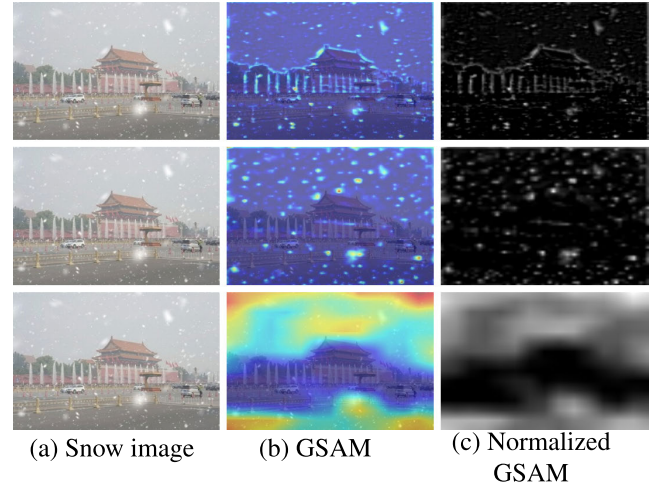


FIGURE 7. Ablation study on visualization on GSAM at different layers.

TABLE 1. Ablation study on component-wise analysis of our method using the SRRS dataset. High values of PSNR and SSIM indicate accurate results.

Method	PSNR $\uparrow$	SSIM $\uparrow$
Baseline	26.58	0.9247
Baseline + GSAM	26.97	0.9253
Baseline + GSEM	27.21	0.9298
Baseline + GSEM + GSAM	28.18	0.9340

TABLE 2. Ablation study on comparison of snow attention mask estimation using the Snow100K dataset.

Metric	PSNR $\uparrow$	SSIM $\uparrow$
DehazeNet [32]	19.622	0.3271
DeepLab [33]	18.8679	0.2942
DesnowNet [4]	22.0054	0.5662
Ours	22.8366	0.6384

Please note that from the image restoration point of view, observed images can be considered as the combination of clean images and noise. For example, in deraining problems, we can model rain images by the combination of clean images and rain streak. If we use (6) for this problem, we can obtain a clean image by subtracting rain streak (snow mask) from the rain image (snow image). Then, we repeatedly perform supervised learning so that differences between clean and ground-truth images decrease. (8) shows the loss function for this supervised learning. In particular, L_r and L_p calculate the difference between two aforementioned images, whereas other losses help estimate the accurate snow masks. Considering shape and irregularity of snow are important to prevent the proposed deep neural network from being dataset-dependent and to boost the generalization performance in real-world scenarios. While real-world snow images severely depend on various shapes and appearances of snow, they are very irregular due to camera specifications, shutter speed, wind, and lighting conditions. Thus, it is important to consider the shape and irregularity of snow.



FIGURE 8. Qualitative comparisons of desnowing results using the SRRS dataset [3].

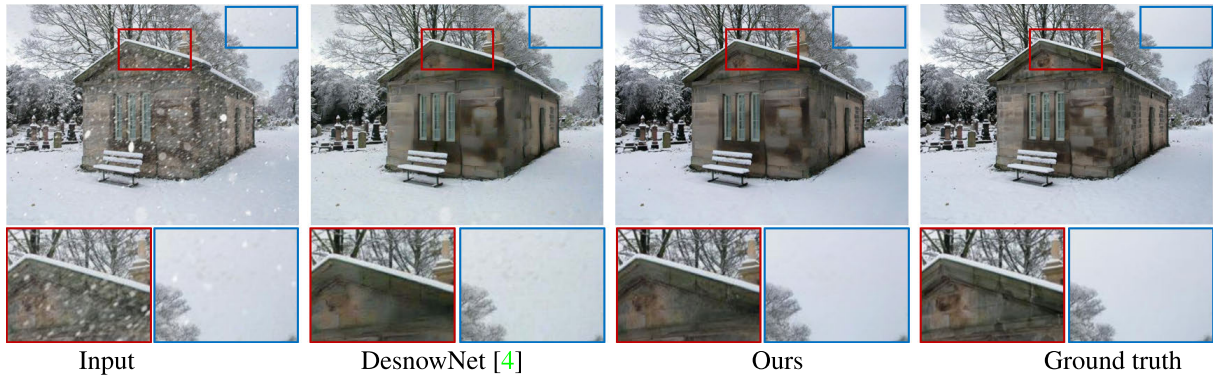


FIGURE 9. Qualitative comparisons of desnowing results using the Snow100K dataset [4]. Boxes contain desnowing results with high-resolution.

IV. EXPERIMENTS

A. IMPLEMENTATION DETAILS

In this section, we describe two training processes in detail. One is for binary classification to estimate the GSAM. Another is for desnowing.

1) BINARY CLASSIFICATION

To estimate the GSAM, we designed a binary classification network based on the VGG16 architecture and pretrained the network using the ImageNet dataset. Then, we trained the network with a batch size of 16 and learning rate of $1e-4$ for 10 epochs using pairs of snow and ground truth images. We extracted the GSAM from the 15-th layer of the network using gradient CAM [6].

2) DESNOWING

Using GSAM and GSEM, we estimated the snow mask and obtained clean images by determining the differences between the input snow images and the estimated masks. For data augmentation, we randomly cropped training images into multiple patches of 240×240 size and flipped the patches horizontally and vertically. For training, we used Adam optimization ($\beta_1 = 0.9$ and $\beta_2 = 0.999$). We trained the network with a learning rate of $2e-4$ for 100k and 300k iterations on the srrs and snow100k dataset each. and an exponential decay ($\gamma = 0.95$) was applied at every 10k iterations after 50k iterations. All training processes were conducted using a single NVIDIA RTX 2080 Ti. Our method was implemented using PyTorch 1.6. To balance the training

loss value, hyperparameters $\{\lambda_i\}_{i=1}^5$ in (8) were set to 10, 1, 1, 10, and 10, respectively.

B. EXPERIMENTAL SETTINGS

We used standard benchmark desnowing datasets including the Snow100K dataset [4] and SRRS dataset [3]. The Snow100K dataset includes 50,000 training images comprising multiple triplets of snow-free, snow synthetic, and snow mask images. In addition, this dataset contains a total of 50,000 test images; there are three types of test images of about 17k each, based on the snow particle size (*i.e.*, small, medium, and large sizes). In the dataset, snow particles are extremely dense under various environments, which makes desnowing challenging. The SRRS dataset [3], the most recent dataset, includes approximately 15,000 synthetic snow images. Similar to the Snow100K dataset, the SRRS dataset consists of small, medium, and large snow particle images. In particular, the veiling effect was synthesized and included in the dataset because snow particles exhibit characteristics of haze if the distance increases. For qualitative evaluation, we used real-world snow images in [4]. The proposed desnowing method runs at approximately 0.05 seconds for an image with a size of 640×480 using a single A6000 GPU.

C. ABLATION STUDY

1) COMPONENT-WISE ANALYSIS

Table 1 summarizes the effectiveness of the proposed GSAM and GSEM. For this experiment, we developed and evaluated three variants of the proposed method using the SRRS dataset [3]: baseline + GSAM, baseline + GSEM, and baseline + GSEM + GSAM. The baseline method used the reconstruction and perceptual losses. Three variants used the snow mask loss for GSAM and the mask edge refine loss for GSEM. We apply to two metrics structural similarity (SSIM) and peak signal-to-noise ratio (PSNR) for evaluation. PSNR is a pixel-wise evaluation method that contains MSE and is the most commonly used metric in image restoration. SSIM measures overall structural similarity. Because MSE could be included in the PSNR formula, we mainly used PSNR to evaluate the desnowing performance as like other low-level vision papers. As shown in Table 1, our method considerably outperforms baselines, and each component of our method contributed in improving the desnowing performance. Our method with both GSEM and GSAM components exhibited the best performance.

2) SNOW MASK ESTIMATION

Figure 6 shows estimated snow attention masks using the Snow100K dataset. As shown in the first and third rows, our method inaccurately estimated masks around sky and clouds areas, which originally had white colors. Nevertheless, our method typically produced accurate snow masks, which were similar to GT masks. Table 2 compares snow masks estimated by different methods using the Snow100K dataset. Although the snow masks were very clutter and difficult to estimate, our

TABLE 3. Quantitative comparison with state-of-the-art desnowing methods on the SRRS datasets. For JSTASR, we showed PSNR and SSIM results in the original paper [3]. In addition, we tested JSTASR using publicly available code provided by authors (*i.e.*, JSTASR*). It should be noted that our method outperforms JSTASR in any case. Standard deviations (STD) were also reported in (-).

Method	PSNR* (STD)	SSIM* (STD)
JSTASR [3]	25.7	0.86
JSTASR*	21.45 (0.0742)	0.76 (0.0018)
Ours	28.27 (0.1116)	0.94 (0.0007)

method produced the most accurate maps in terms of PSNR and SSIM.

3) VISUALIZATION OF GSAM

Figure 7 shows the gradient based snow activation map (GSAM) for several layers of VGG16. The first row of Figure 7 illustrates the GSAM for the maxpool layer 1 (*i.e.*, 5th layer), the second row represents the GSAM for the conv 3-3 layer (*i.e.*, 15th layer), and the third row describes the GSAM for the conv 5-1 layer (*i.e.*, 25th layer). The first, second, and third columns of Figure 7 contain snow images, color maps of the GSAM, and normalized GSAMs, respectively. As shown in the first row, our method accurately estimated snow positions, however it wrongly emphasized object edges. In the third row, the method estimated snow areas globally, thus it was difficult to recognize the exact snow positions. In contrast, in the second row, our method most accurately estimated snow areas. Thus, we used the conv 3-3 layer to obtain the proposed GSAM.

D. COMPARISON WITH OTHER METHODS

We compared our method with recent state-of-the-art methods (*i.e.*, JSTASR [3] and DesnowNet [4]) using Snow100K and SRRS datasets. In addition, we also conducted experiments on real-world snow images.

1) SYNTHETIC DATASET

In JSTASR [3], training and testing sets need to be selected randomly, but experimental settings were not exactly described in [3]. Thus, we could not know selected train and test sets used in [3]. Thus, for fair comparisons, we tested 100 times by randomly selecting 1000 test images that are not included in our training set to follow a uniform distribution. Then, we reported averaged PSNR and SSIM values (PSNR* and SSIM*, respectively) with standard deviation, as shown in Table 3. In addition, we conducted exhaustive experiments by considering two cases.

- Case 1: We used PSNR and SSIM results of JSTASR in the original paper [3].
- Case 2: We tested JSTASR using publicly available code provided by authors (*i.e.*, JSTASR*).

As shown in the table, our method outperforms JSTASR in any case. The proposed method considerably outperforms the state-of-the-art method, JSTASR, in terms of PSNR and SSIM. Figure 8 shows qualitative desnowing results

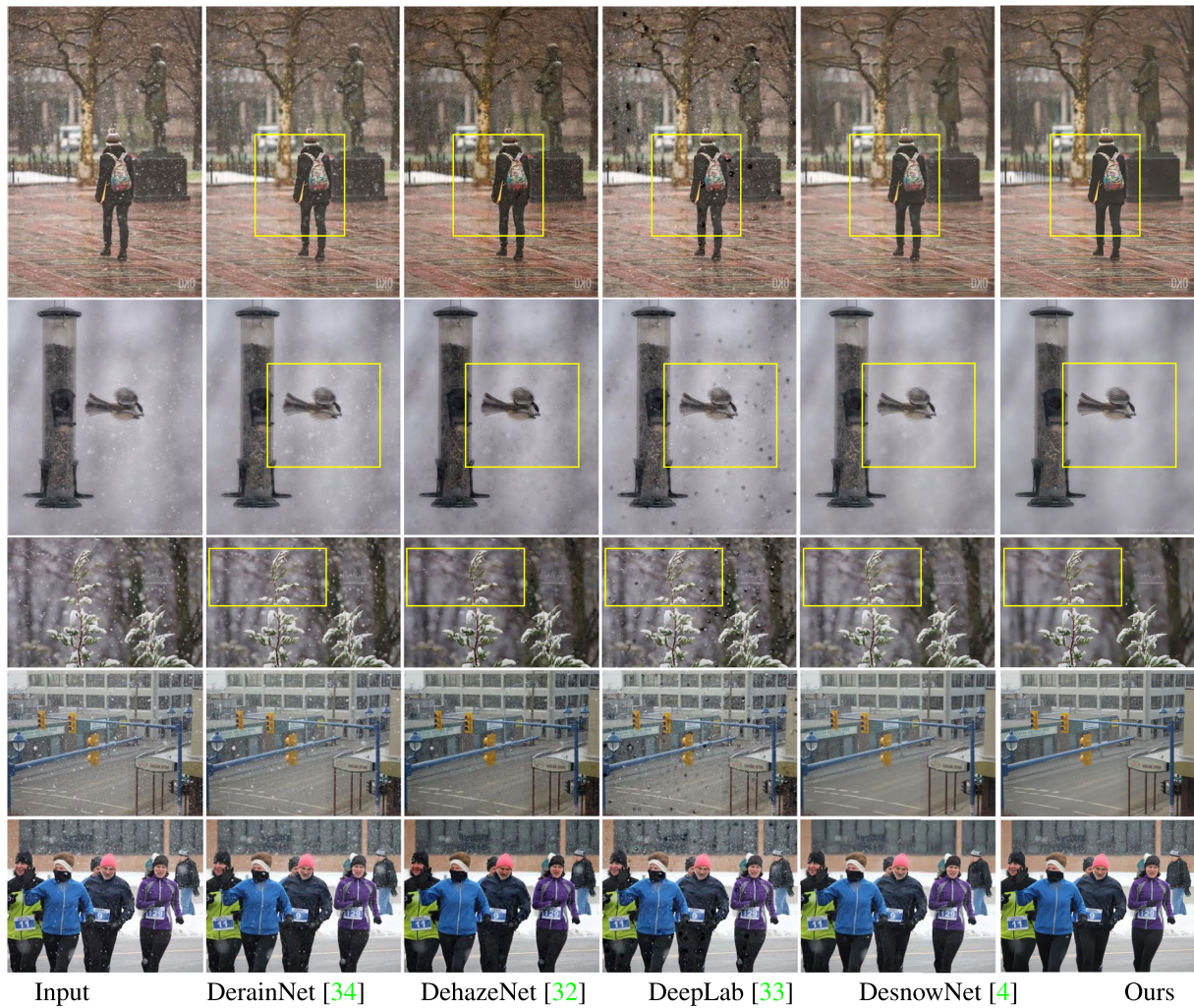


FIGURE 10. Desnowing results for real-world images. Yellow boxes contain desnowing results with high-resolution.

TABLE 4. Quantitative comparison with state-of-the-art desnowing methods on the Snow100k dataset. Red and blue numbers denote the best and the second best results, respectively.

Subset	Snow100K-S		Snow100K-M		Snow100K-L	
Metric	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
DerainNet [34]	25.7457	0.8699	23.3669	0.8453	19.1831	0.7495
DehazeNet [32]	24.9605	0.8832	24.1646	0.8666	22.6175	0.7975
DeepLab [33]	25.9472	0.8783	24.3677	0.8572	21.2931	0.7747
JORDER [5]	25.6202	0.8861	24.974	0.8716	23.4085	0.8091
Desnownet [4]	32.3331	0.95	30.8682	0.9409	27.1699	0.8983
All in One [35]	-	-	-	-	28.33	0.8820
SMGARN [17]	34.46	0.95	33.22	0.95	29.44	0.93
Ours	34.5051	0.9458	32.7869	0.9345	28.7576	0.8843

on the SRRS dataset. Our method shows state-of-the-art performance. While JSTASR remained stains where snow is located, our method produced accurate desnowing images.

Table 4 shows quantitative comparisons of desnowing results on the Snow100K dataset. DerainNet [34], DehazeNet [32], DeepLab [33], and JORDER [5] were reimplemented by [4]. SMGARN [17] is the state-of-the-art

method for desnowing. For this experiment, we used Snow100K-S, M, and L testsets. As shown in Table 4, DesnowNet typically presented accurate desnowing images, especially in terms of SSIM. All in One [35] also produced promising results in terms of PSNR for the Snow100K-L dataset, while it could process several tasks at the same time using neural architecture search. The proposed method accurately removed snow particles from snow images.

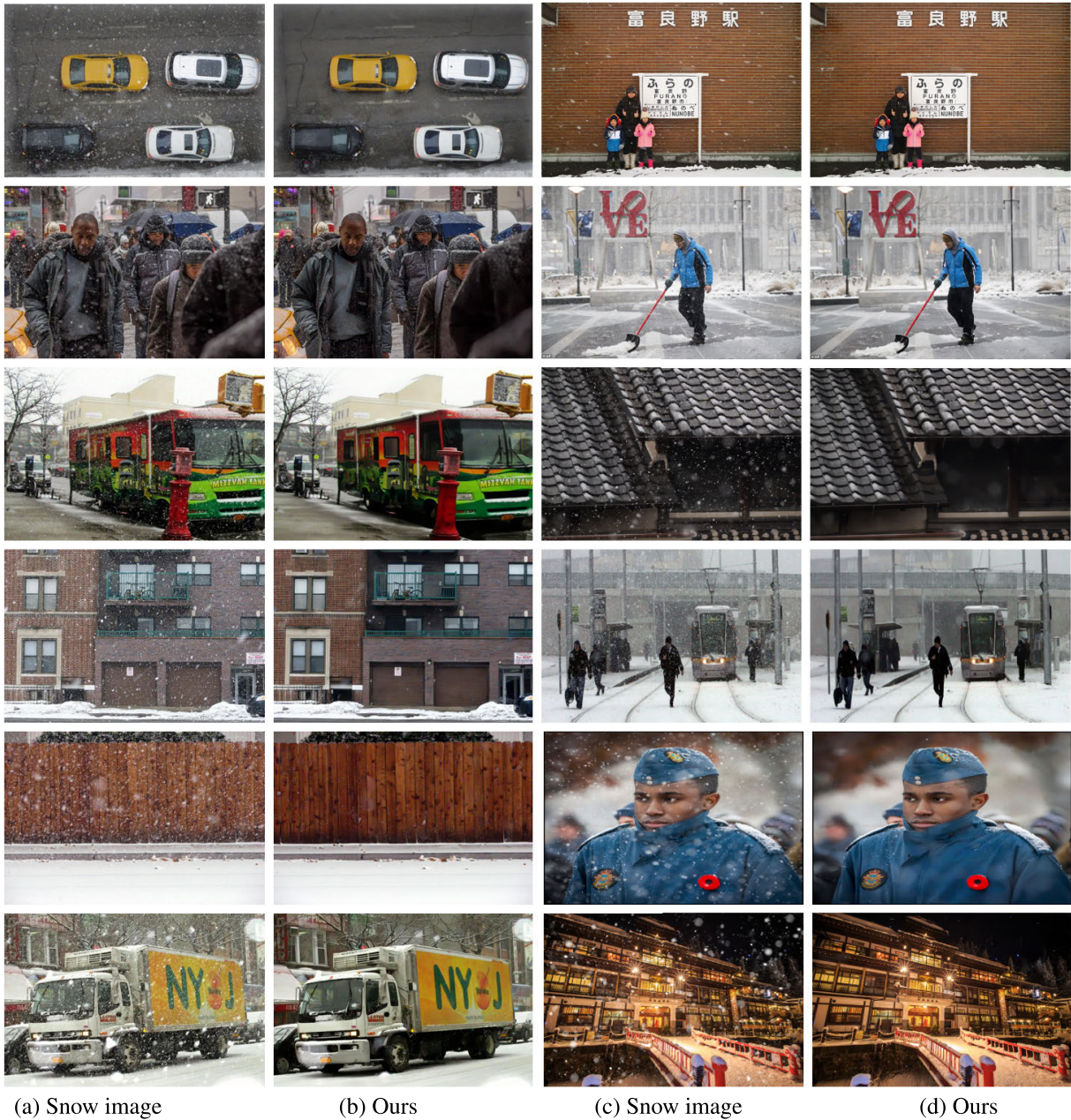


FIGURE 11. Qualitative results on the real-world datasets.

SMGARN [17] showed the best performance using advanced snow masks. For all datasets, our method was comparable to the state-of-the-art desnowing method, SMGARN in terms of the PSNR. Figure 9 contains a heavy snow image in the Snow100K dataset, in which desnowing is very difficult. DesnowNet produced artifacts in the blue box, whereas the proposed method exhibited considerably clean results. As shown in areas with red boxes, our method more accurately described image details than DesnowNet. Please note that SMGARN is the latest method, thus qualitative evaluation has not been conducted because the corresponding trained model has not been provided publicly.

2) REAL-WORLD DATASET

For real-world image evaluation, we used the Snow100K dataset. Figure 10 shows qualitative desnowing results for real-world snow images in the Snow100K dataset. DerainNet and DehazeNet could not accurately remove snow particles. DeepLab could not restore context information on the areas where snow particles were removed and contained several artifacts in images. The proposed method and DesnowNet exhibited high-quality snow particle removal results. However, DesnowNet produced many faint snow particles in the background, whereas the proposed method smoothly removed these faint snow particles, as shown in



FIGURE 12. Detection results in real-world images under heavy snow environments.

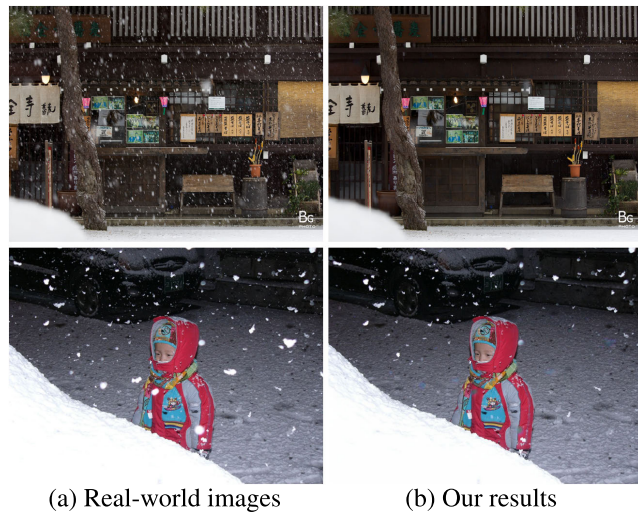


FIGURE 13. Failure examples on real-world snow images, which have very small particles and precise particles without blurring.

the yellow box of Figure 10. DesnowNet and our method produced accurate desnowing results in the fourth and fifth rows. We present additional qualitative results of our method for real world datasets. Figure 11 shows qualitative results on the real world dataset and demonstrate that our method accurately removed snow particles in real-world snowing scenarios.

E. DESNOWING FOR HIGH-LEVEL VISION TASKS

Desnowing can be used as a pre-processing method for high-level vision tasks in outdoor conditions. In this experiment, we used the proposed desnowing method as a pre-processing step for object detection tasks, because object detection performance is highly affected by weather conditions. We used DETR [36] for the object detector. Figure 12 shows object detection results in real-world images under snow environments. The first row contains the object detection

results without any processing step, whereas the second row shows the object detection results after the desnowing process. If our desnowing method is not applied as a preprocessing step, the object detection results was poor. For example, a person and a truck, the most noticeable objects in the first and second columns, were not detected. In the third column, the tent in the center was mistakenly recognized as a bus. In contrast, our desnowing method accurately removed snowfalls and improved the detection performance, which enabled to detect objects successfully.

V. DISCUSSION

Desnowing has difficulties in constructing real-world datasets, because it is difficult to capture real snow images. Thus, conventional datasets such as Snow100K [4] and SRRS [3] synthesize snow images. Snow100K is a large-scaled dataset of 50,000 images and synthesizes snow particles with various sizes. However, this dataset does not reflect the climate of snowy weather, thus contains erroneous images with snow particles under the sunny weather SRRS synthesizes snow particles by adopting similar techniques used in the haze dataset. Thus, it can consider the veiling effect, which can occur in snowfall images.

Figure 13 shows failure cases in real-world snow images. The proposed method produced state-of-the-art desnowing results without using additional information such as masks in Tables 3 and 4. As shown in the first row, the method also showed good desnowing performance in real-world snow images. However, very small particles remained on the right side and all small particles were not clearly removed, because the method estimated inaccurate gradients of snow particles. The second row images include snow particles without any shaking and blurring. Very precisely expressed particles were not clearly removed by our method, although most snow areas were removed to some extent. Please note that conventional

datasets (*i.e.*, Snow100K and SRRS) did not contain the aforementioned type of snow particles.

Comparing other image restoration tasks, there are relatively few works on desnowing because it is nontrivial to capture the dynamics of snowing. Thus, we researched desnowing problems and verified desnowing performance via supervised learning. In future work, we will consider unsupervised learning for real-world snow scenarios, in which ground-truths are not available. In addition, because it is difficult to handle all cases of real-world images with only synthetic datasets, we plan to apply unsupervised learning (*i.e.*, domain adaptation) for generalization.

VI. CONCLUSION

In this paper, we proposed a novel dual gradient-based desnowing method. We accurately localized snow particles in an image using the proposed gradient-based snow activation map and recognized various patterns in the shapes and trajectories of snow particles using the proposed gradient-based snow edge map. Using these two maps, we estimated the snow attention mask for snow removal. In addition, a translucency-aware context restoration network was proposed to handle snow translucency and to prevent desnowed images from losing image context information, which may occur during desnowing. Experimental results demonstrate that our method outperforms state-of-the-art desnowing algorithms. Furthermore, we will extend our proposed method to be applicable to other low-level vision tasks such as deraining by taking into account the common characteristics of different low-level vision problems.

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